IMD0904 –Plataforma de Hardware para Internet das Coisas

Prof. Daniel Lopes Martins

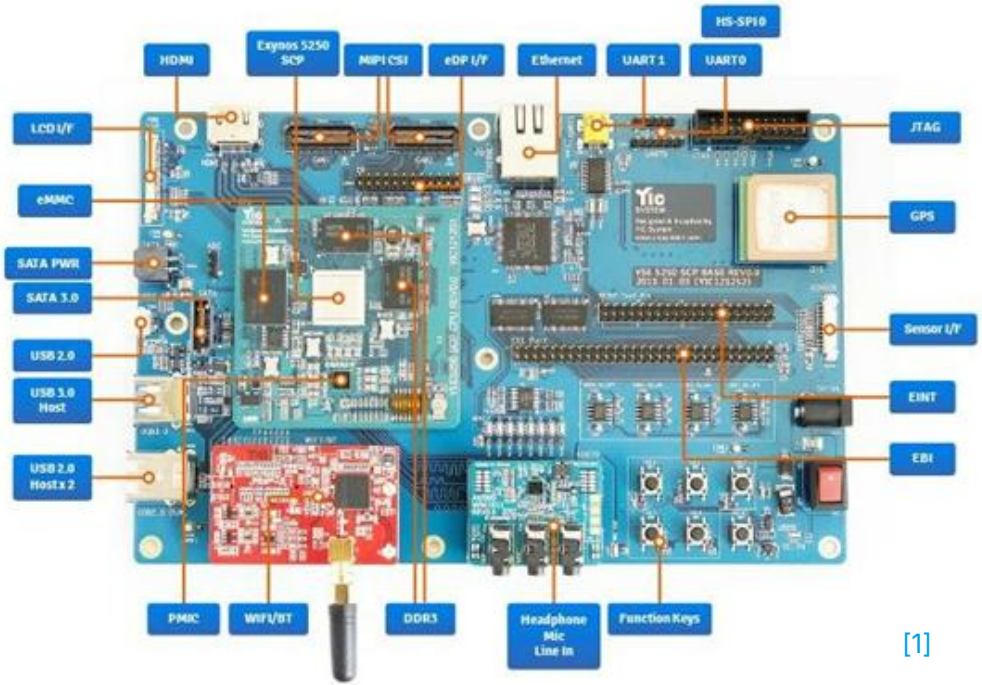
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AULA 04

Conhecendo o Arduino e o ESP32

Plataformas de Hardware

- Uma plataforma de hardware é um conjunto de componentes físicos (hardware) projetado para executar um determinado sistema ou aplicação.

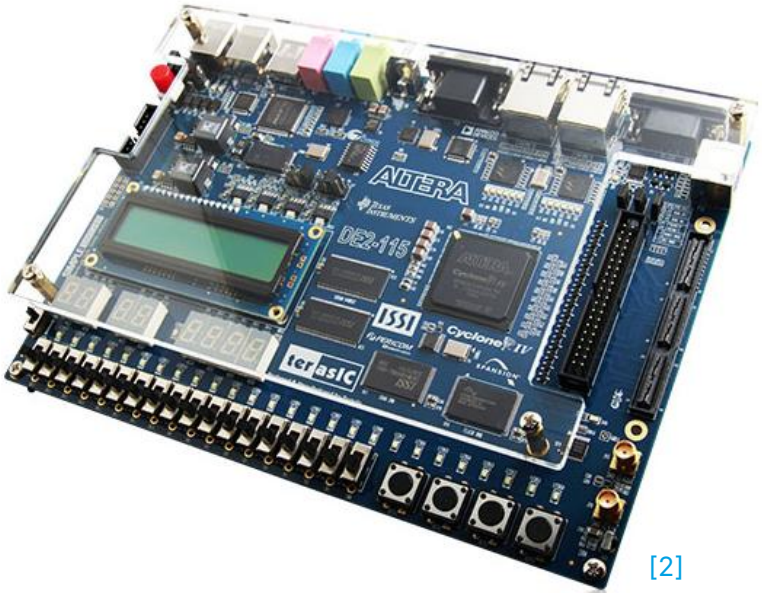


[1]

Plataformas de Hardware

- Essas plataformas incluem:
  - Processadores.
  - Interfaces de comunicação.
  - Memórias.
  - Sensores e atuadores.

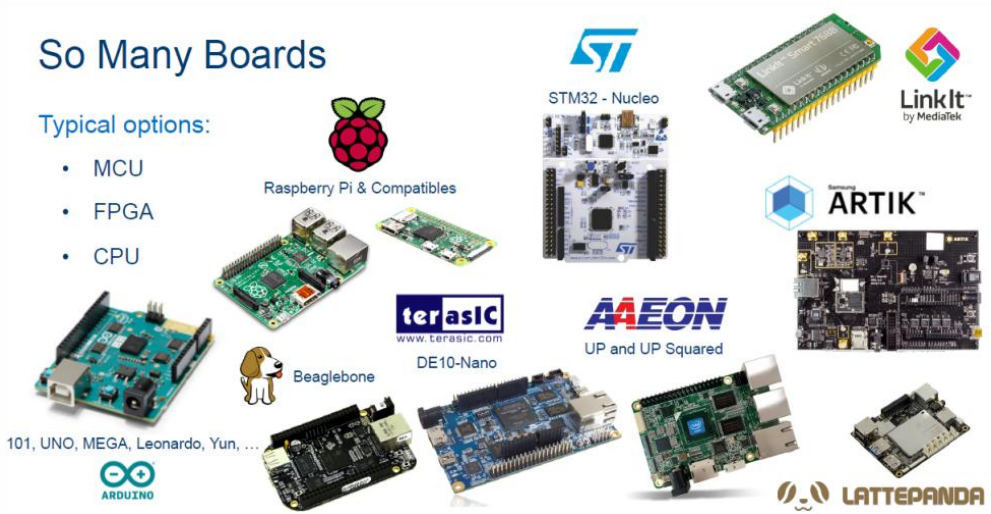
- Permitem desenvolvimento de soluções embarcadas e conectadas.



[2]

Plataformas de Hardware

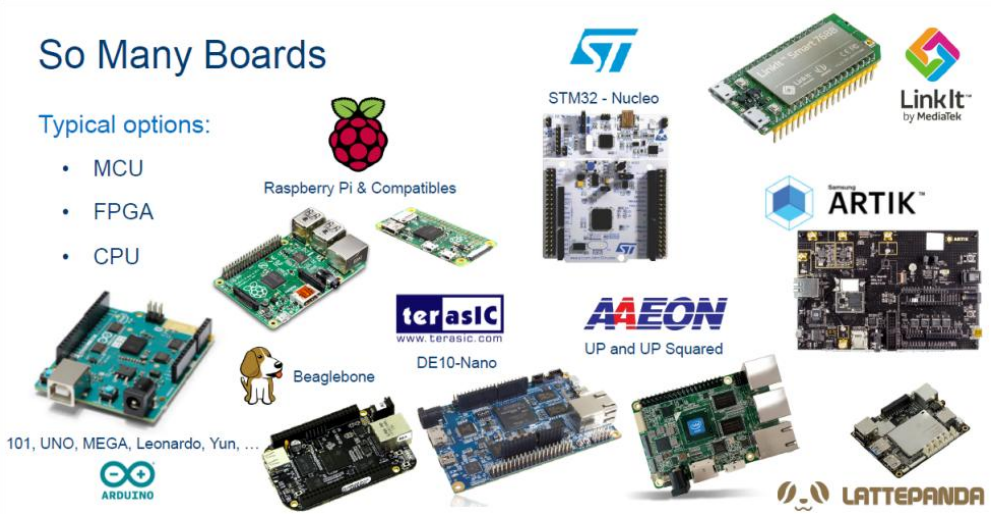
- No contexto de IoT:
  - Dispositivo ou conjunto de dispositivos físicos capazes de coletar, processar e transmitir dados para a rede.
- Possibilitam a automação e o controle remoto de sistemas.



[3]

Plataformas de Hardware

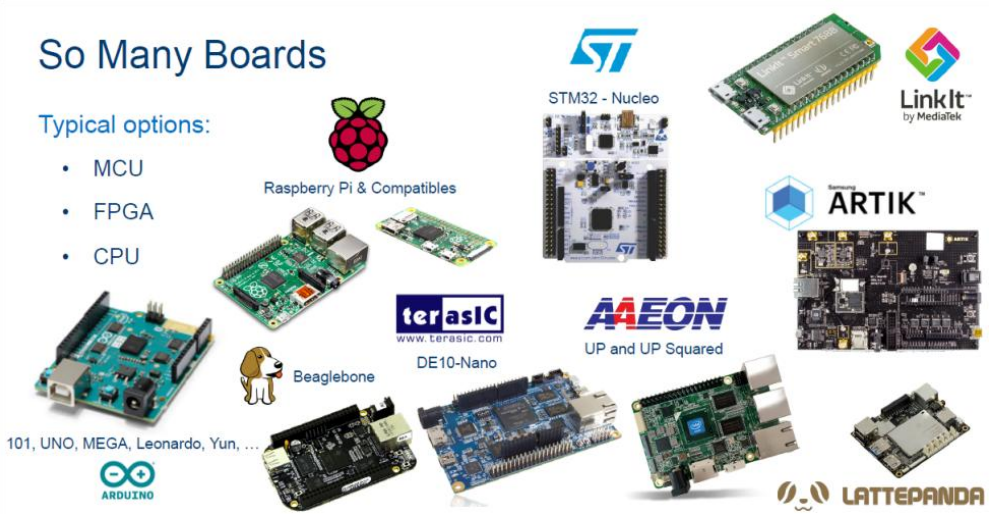
- Exemplos:
  - MCUs:
    - Arduino, ESP32, STM32, Raspberry Pi Pico.
  - SBCs (CPU):
    - Raspberry Pi, BeagleBone Black, Odroid N2+.
    - NVIDIA Jetson, Coral Edge TPU, Altera DE10-Nano.



[3]

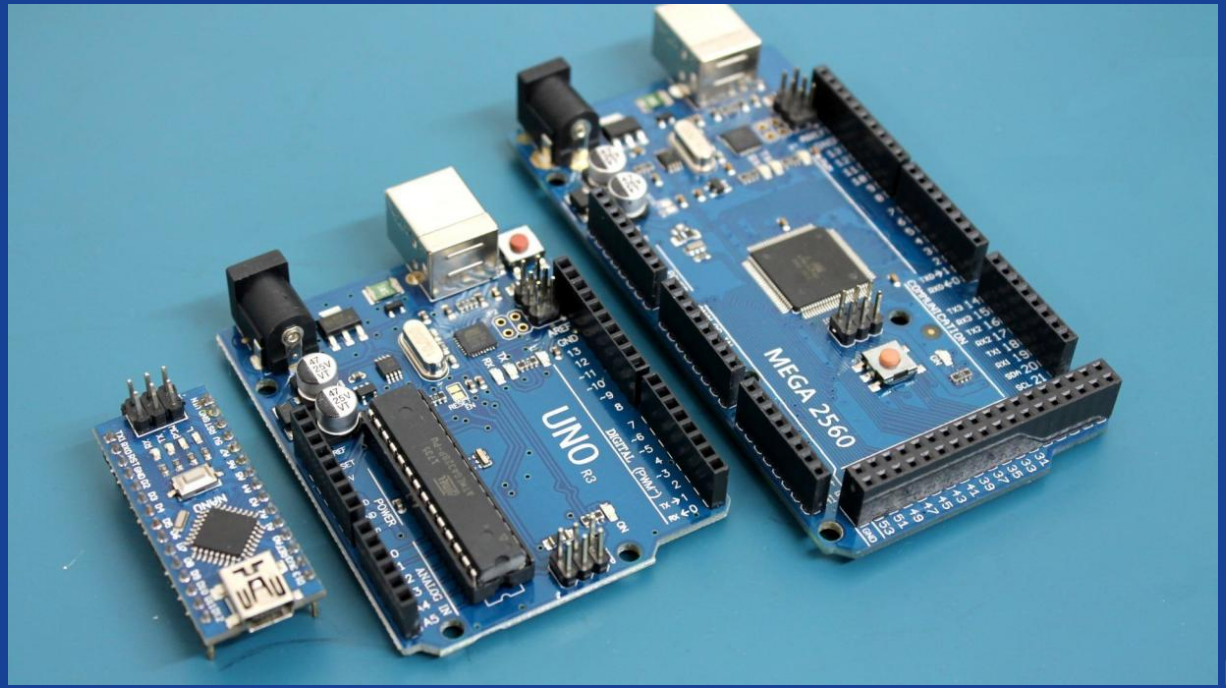
Plataformas de Hardware

- Exemplos:
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    - NVIDIA Jetson, Coral Edge TPU, Altera DE10-Nano.



[3]

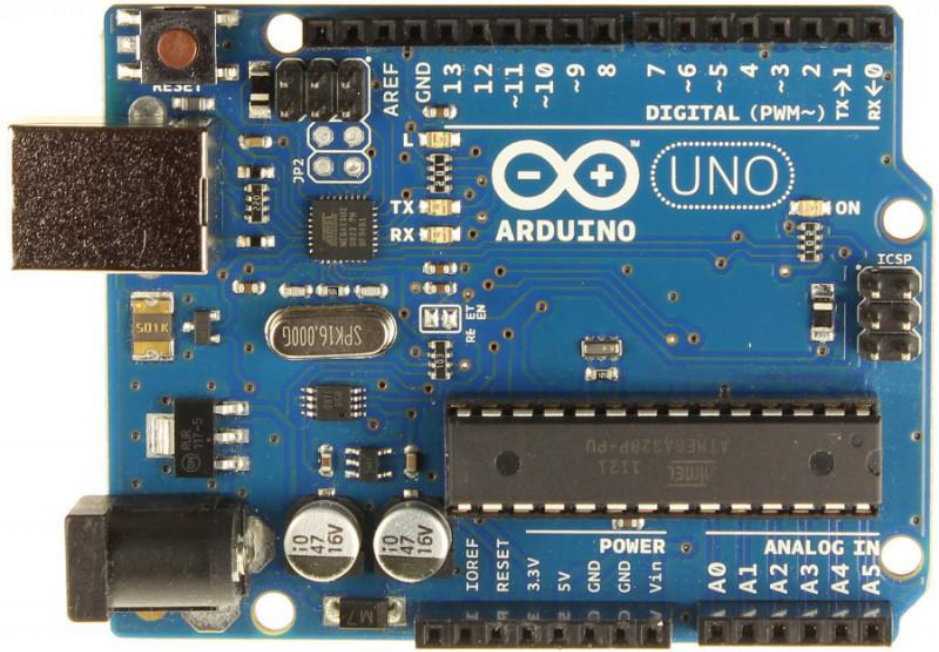
PLATAFORMA ARDUINO



[4]

Plataforma Arduino

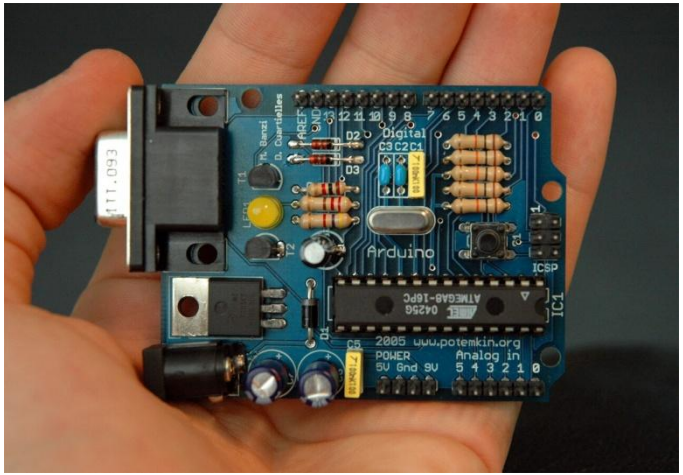
- O que seria o Arduino?
  - Plataforma de hardware e software de código aberto.
  - Baseado em microcontroladores programáveis (megaAVR, ESP32, ARM Cortex-M).
  - Ambiente de desenvolvimento próprio (Arduino IDE).



[5]

Plataforma Arduino

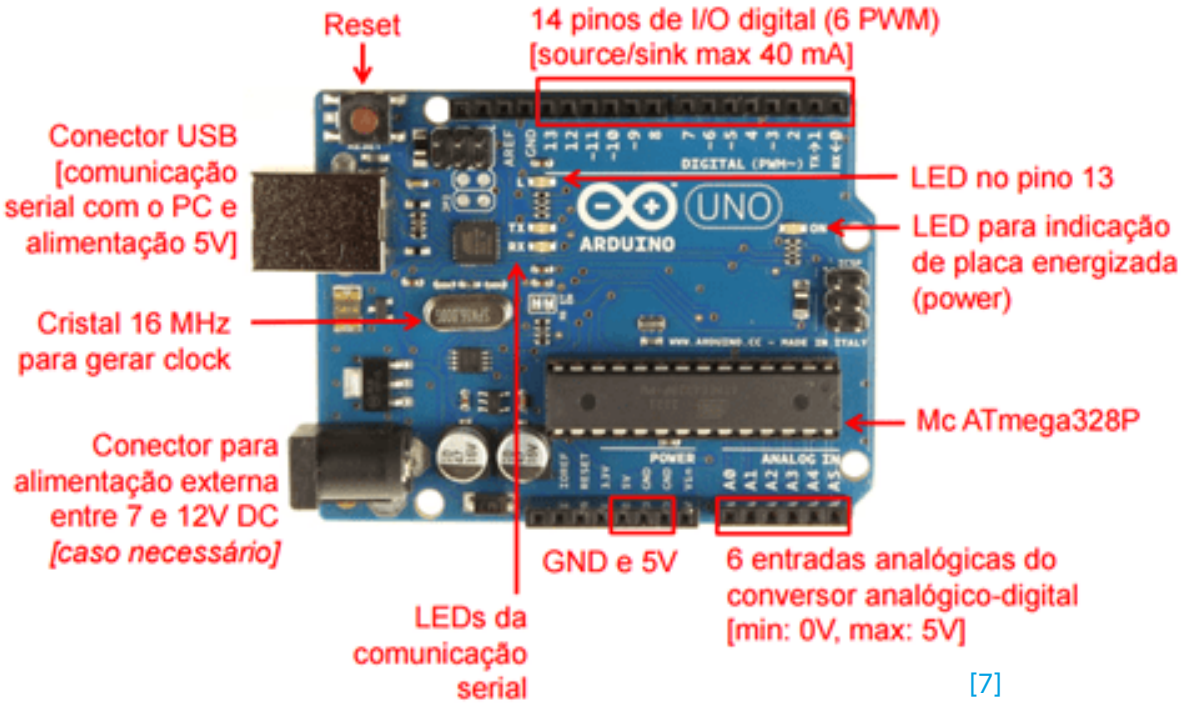
- O que seria o Arduino?
  - História
    - Projeto Wiring (2003 – 2005).
    - Criação do Arduino (2005).
    - Expansão e popularização (2006-2015).
    - Conflito e Divisão (2015 – 2017).
    - Fase atual (2018 – presente).



[6]

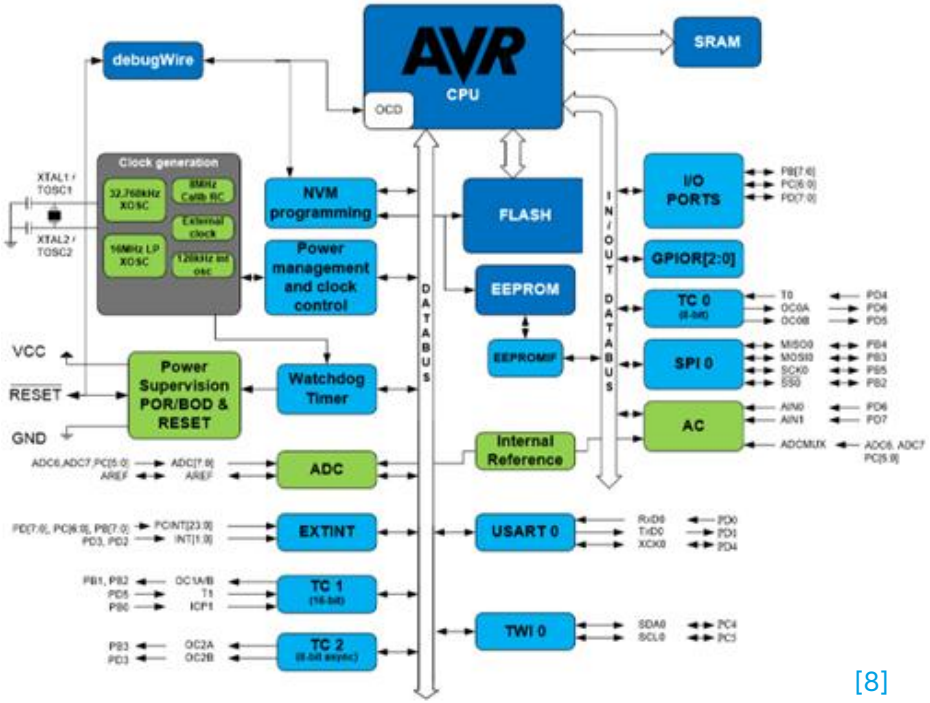
Plataforma Arduino

- Características:

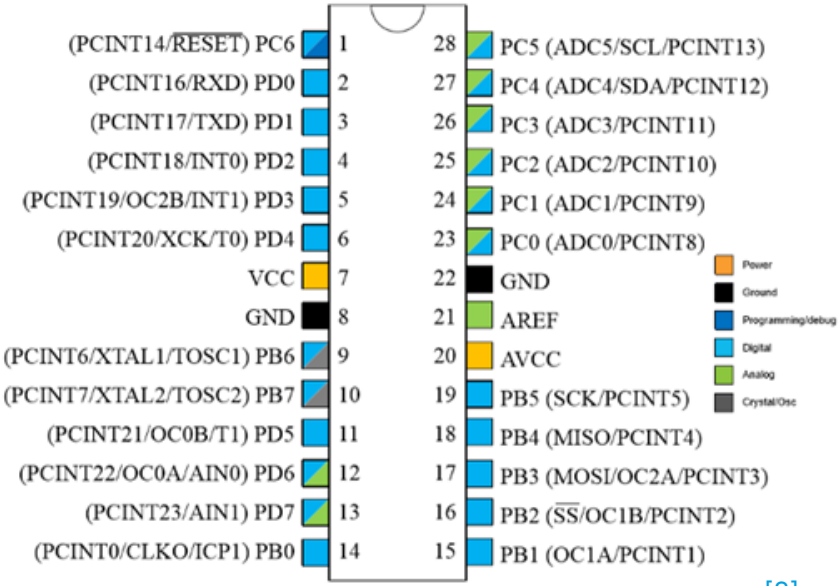


[7]

- Características:



[8]



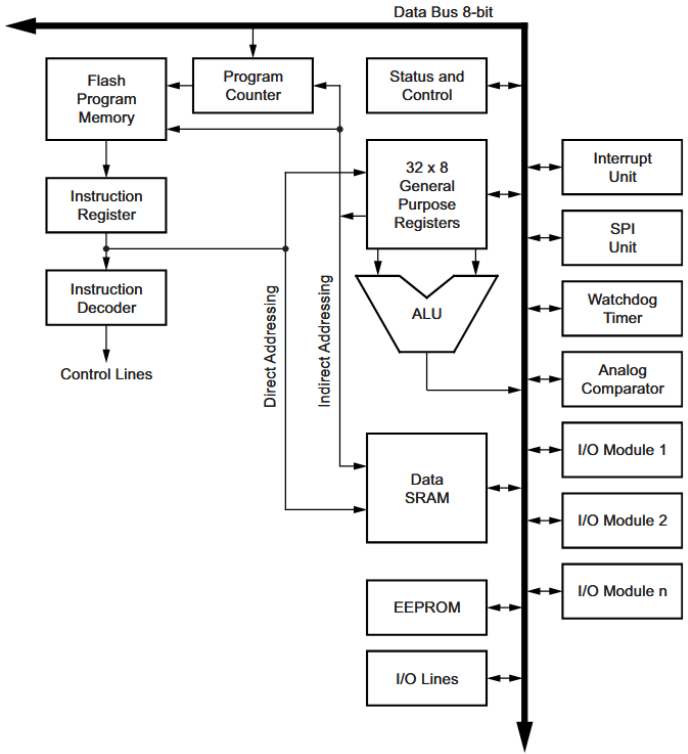
[8]

Plataforma Arduino

- Características:
  - Tipo de arquitetura: AVR 8bit RISC.
  - 131 instruções, a maioria de 1 ciclo de clock.
  - Clock: 20 MHz (alguns modelo, 16 MHz UNO por exemplo).
  - Memória: 32 KB Flash, 2 KB SRAM, 1 KB EEPROM.
  - IO: 23 digitais (6 saídas PWM), 6 ADC (10 bits), 1 UART, 1 I2C, 1 SPI.
  - Timers: 3 (1 de 16 bits e 2 de 8 bits).
- Onde encontrar essas características? Datasheet.

- Características:

Figure 6-1. Block Diagram of the AVR Architecture



[9]

- Características:

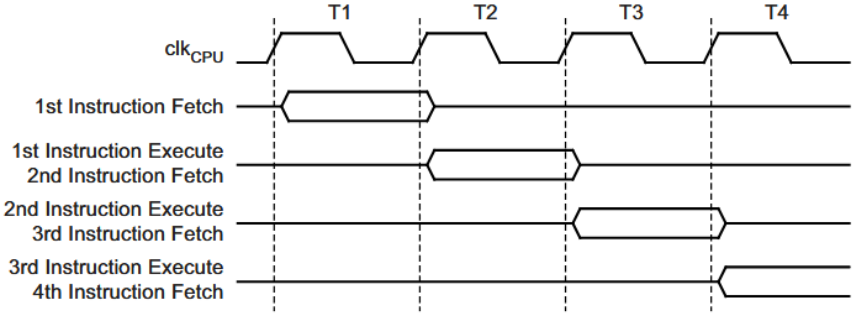
31. Instruction Set Summary

Mnemonics	Operands	Description	Operation	Flags	#Clocks
Arithmetic and Logic Instructions					
ADD	Rd, Rr	Add two registers	Rd ← Rd + Rr	Z,C,N,VH	1
ADC	Rd, Rr	Add with carry two registers	Rd ← Rd + Rr + C	Z,C,N,V,H	1
ADIW	Rdl, K	Add immediate to word	Rdh: Rdl ← Rdh: Rdl + K	Z,C,N,V,S	2
SUB	Rd, Rr	Subtract two registers	Rd ← Rd – Rr	Z,C,N,V,H	1
SUBI	Rd, K	Subtract constant from register	Rd ← Rd – K	Z,C,N,V,H	1
SBC	Rd, Rr	Subtract with carry two registers	Rd ← Rd – Rr – C	Z,C,N,V,H	1
SBCI	Rd, K	Subtract with carry constant from reg.	Rd ← Rd – K – C	Z,C,N,V,H	1
SBIW	Rdl, K	Subtract immediate from word	Rdh: Rdl ← Rdh: Rdl – K	Z,C,N,V,S	2
AND	Rd, Rr	Logical AND registers	Rd ← Rd × Rr	Z,N,V	1
ANDI	Rd, K	Logical AND register and constant	Rd ← Rd × K	Z,N,V	1
OR	Rd, Rr	Logical OR registers	Rd ← Rd v Rr	Z,N,V	1
ORI	Rd, K	Logical OR register and constant	Rd ← Rd v K	Z,N,V	1
EOR	Rd, Rr	Exclusive OR registers	Rd ← Rd ⊕ Rr	Z,N,V	1
COM	Rd	One's complement	Rd ← 0xFF – Rd	Z,C,N,V	1
NEG	Rd	Two's complement	Rd ← 0x00 – Rd	Z,C,N,V,H	1
SBR	Rd, K	Set bit(s) in register	Rd ← Rd v K	Z,N,V	1
CBR	Rd, K	Clear bit(s) in register	Rd ← Rd × (0xFF – K)	Z,N,V	1
INC	Rd	Increment	Rd ← Rd + 1	Z,N,V	1
DEC	Rd	Decrement	Rd ← Rd – 1	Z,N,V	1
TST	Rd	Test for zero or minus	Rd ← Rd × Rd	Z,N,V	1
CLR	Rd	Clear register	Rd ← Rd ⊕ Rd	Z,N,V	1

[9]

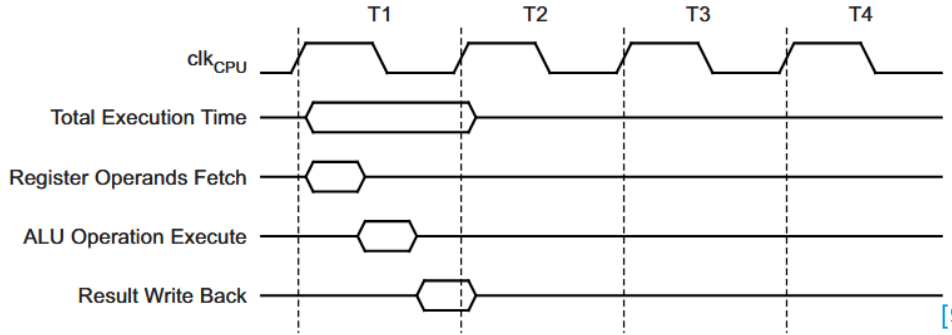
- Características:

Figure 6-4. The Parallel Instruction Fetches and Instruction Executions



[9]

Figure 6-5. Single Cycle ALU Operation

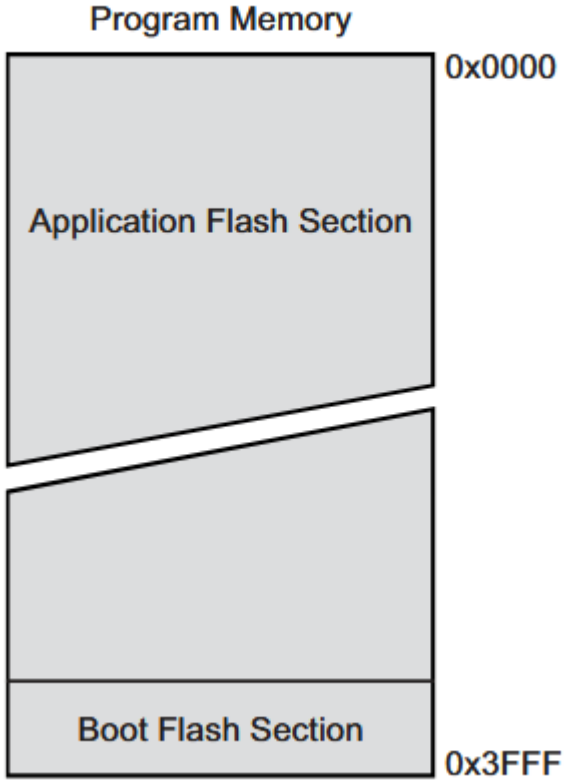


[9]

Plataforma Arduino

- Características:
  - PC: 14bits

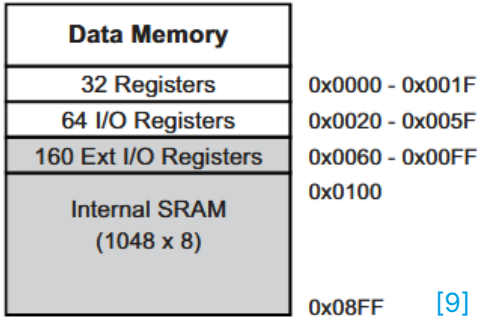
Figure 7-1. Program Memory Map ATmega328P



[9]

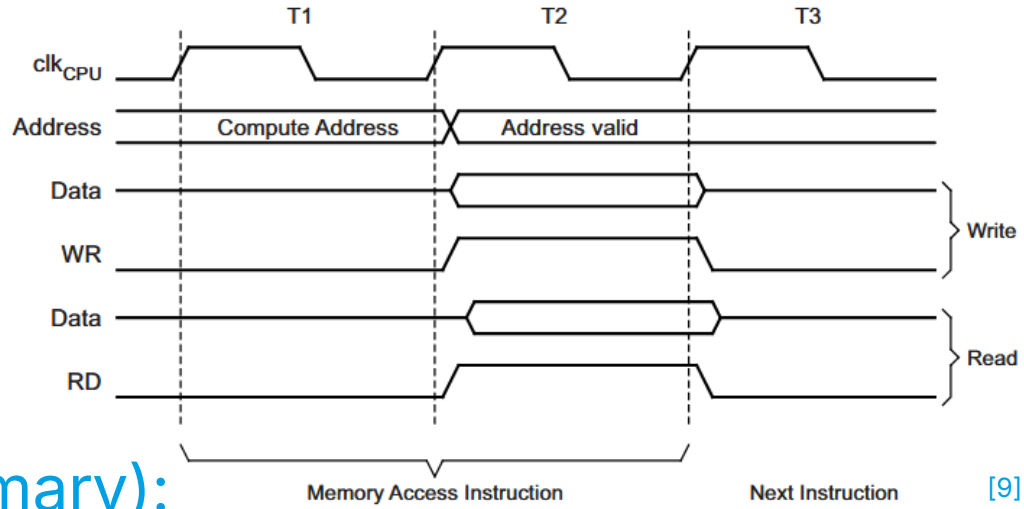
- Características:

Figure 7-2. Data Memory Map



[9]

Figure 7-3. On-chip Data SRAM Access Cycles



[9]

- Exemplo (Register Summary):
  - DDRB – 0x24 – Define direção: 0 = entrada, 1 = saída.
  - PORTB – 0x25 – Define nível lógico dos pinos configurados como saída; ativa pull-up se entrada.
  - PINB – 0x23 – Lê o estado lógico atual dos pinos.

- Exemplo de código em assembly:

```
; Configurar PB2 como saída
    sbr DDRB, 2          ; Set Bit no registrador DDRB (bit 2)
    → saída

; Acender LED (colocar PB2 em nível alto)
    sbr PORTB, 2         ; Set Bit no registrador PORTB (bit 2)
    → HIGH
```

Plataforma Arduino

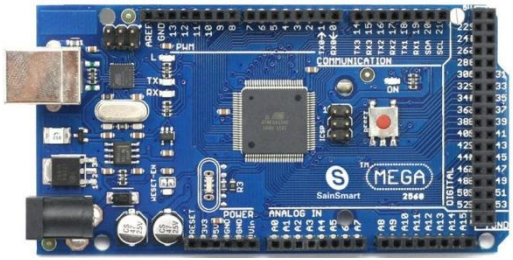
- Modelos conhecidos:
  - 8 bits: Uno, Mega2560, Nano, Micro e Leonardo.
  - Iot e Wifi: Uno WiFi, MKR WiFi1010.
  - Alto desempenho: Nano 33, Due, ESP32, Portenta H7.



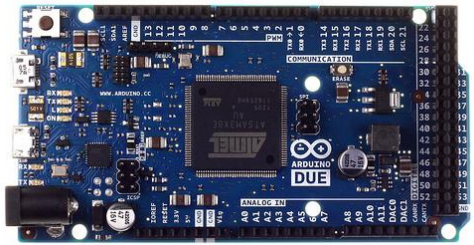
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[10]



[11]



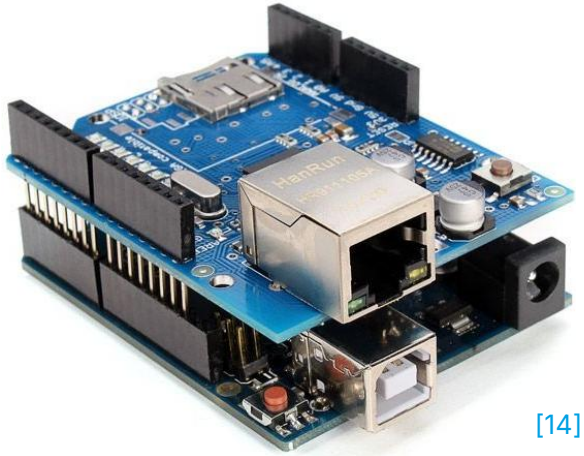
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[13]

Plataforma Arduino

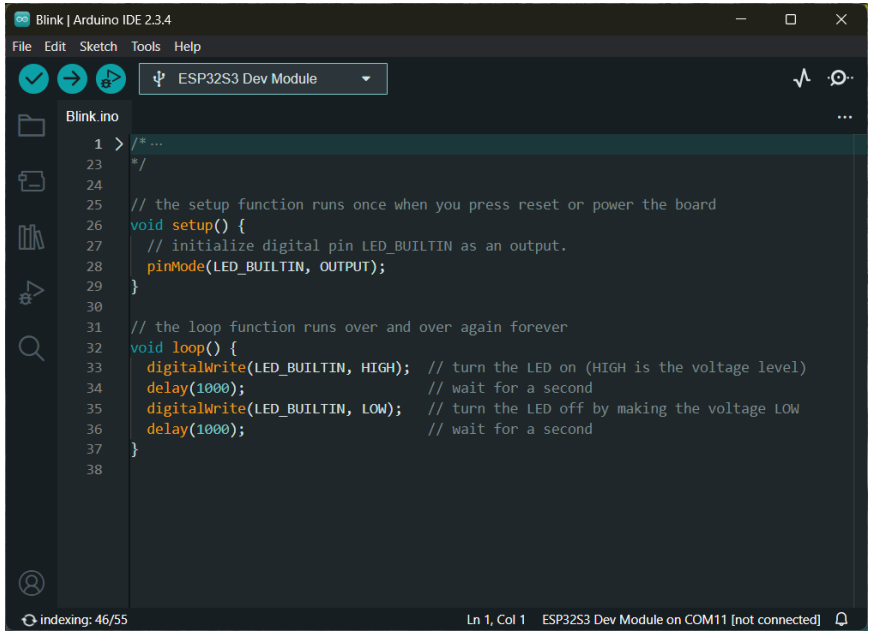
- Shields e módulos
  - O arduino em si tem várias limitações.
  - Criadores e comunidade passaram a desenvolver os shields e módulos.
  - Placas que expandem as funções originais:
    - Ethernet, Motor, WiFi, Sensores, displays.
  - Comunicação: SPI, I^2C (ou I2C), UART.



[14]

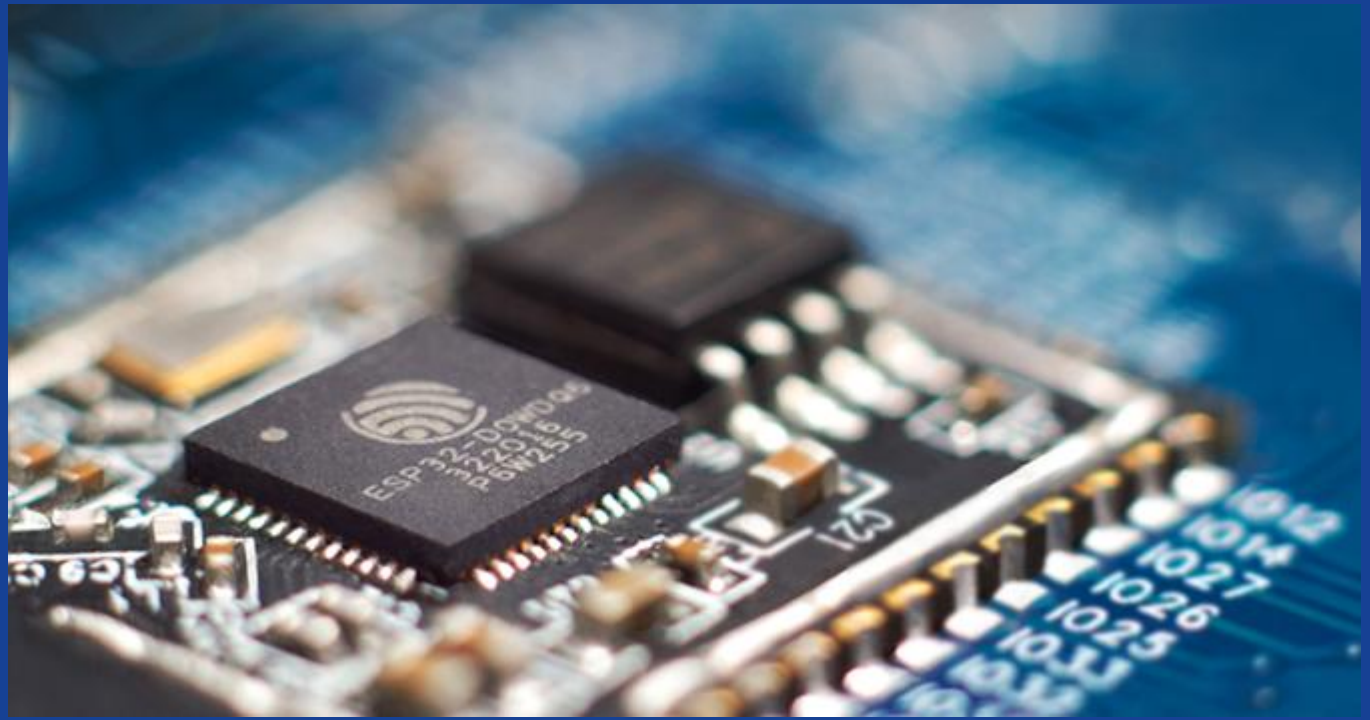
Plataforma Arduino

- Programação
  - IDE Arduino
    - Permite uso de bibliotecas instaladas internamente.
    - Permite instalar modelos de placas diferentes.
    - Linguagem baseada no C/C++.
  - Alternativas
    - PlatformIO (VSCode, Eclipse).



[15]

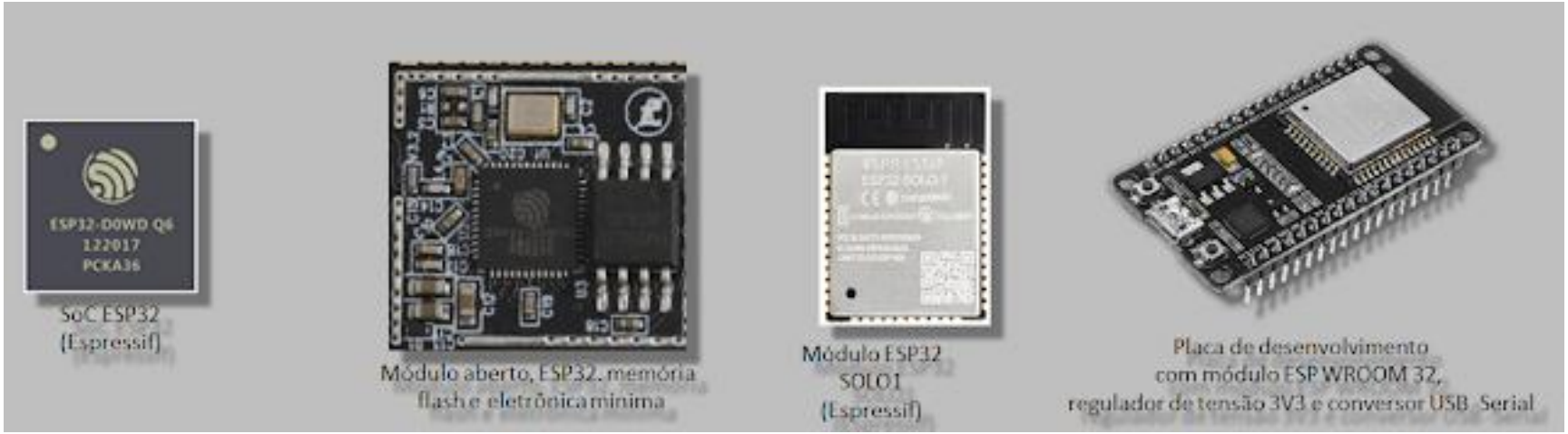
PLATAFORMA ESP32



[16]

Plataforma ESP32

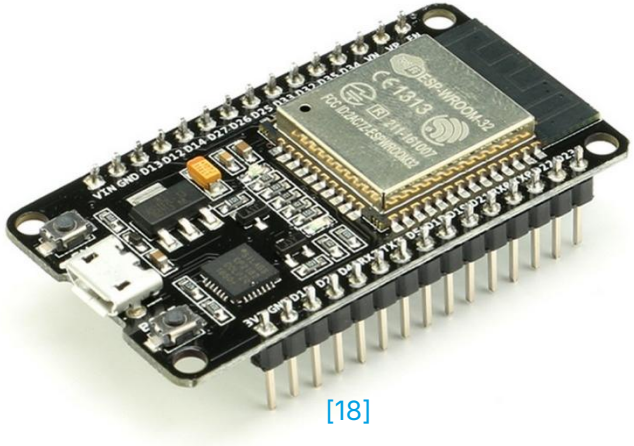
- O que seria o ESP32?
  - É um microcontrolador, mas existem os módulos e diversas plataformas de desenvolvimento que o embarcam.



[17]

Plataforma ESP32

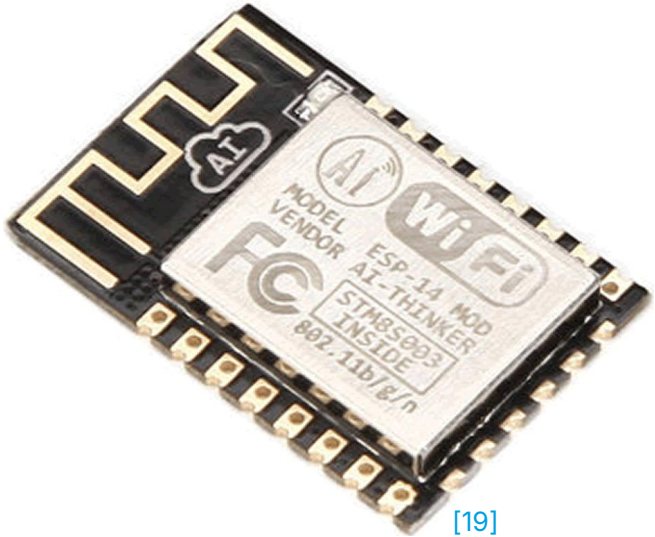
- O que seriam as placas de desenvolvimento ESP32?
  - Baseado no µC ESP32, são parcialmente abertas.
  - Possui o ESP-IDF, framework oficial, mas pode ser programado pelo Arduino IDE e MicroPython (totalmente aberto).



[18]

Plataforma ESP32

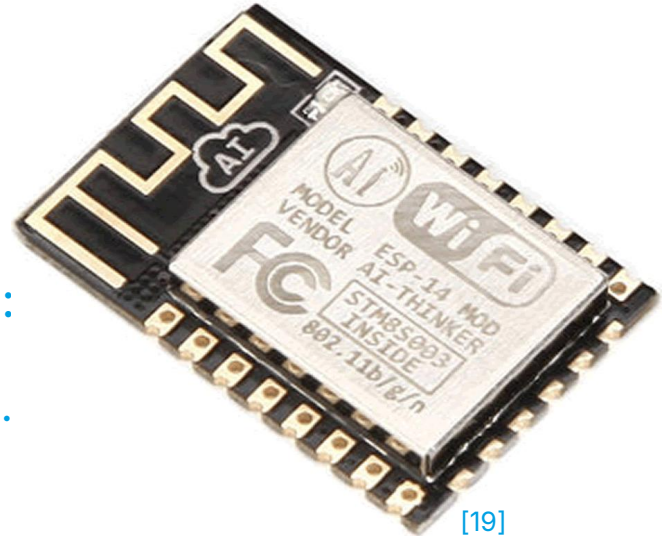
- História
  - ESP8266 (2014)
    - Lançado como chip WiFi barato (WiFi 4) e eficiente.
    - Mas possui uC poderoso embutido.
    - Single core Tensilica L106 (80 MHz).
    - Sem Bluetooth.
    - 160 KB RAM.
    - Firmwares customizados (NodeMCU).
  - ESP32 (2016)
    - Dual Core, Xtensa LX6 (240 MHz), com melhorias no WiFi mais estável (4.0), BLE e classic (v4.2) e mais memória (520KB SRAM).



[19]

Plataforma ESP32

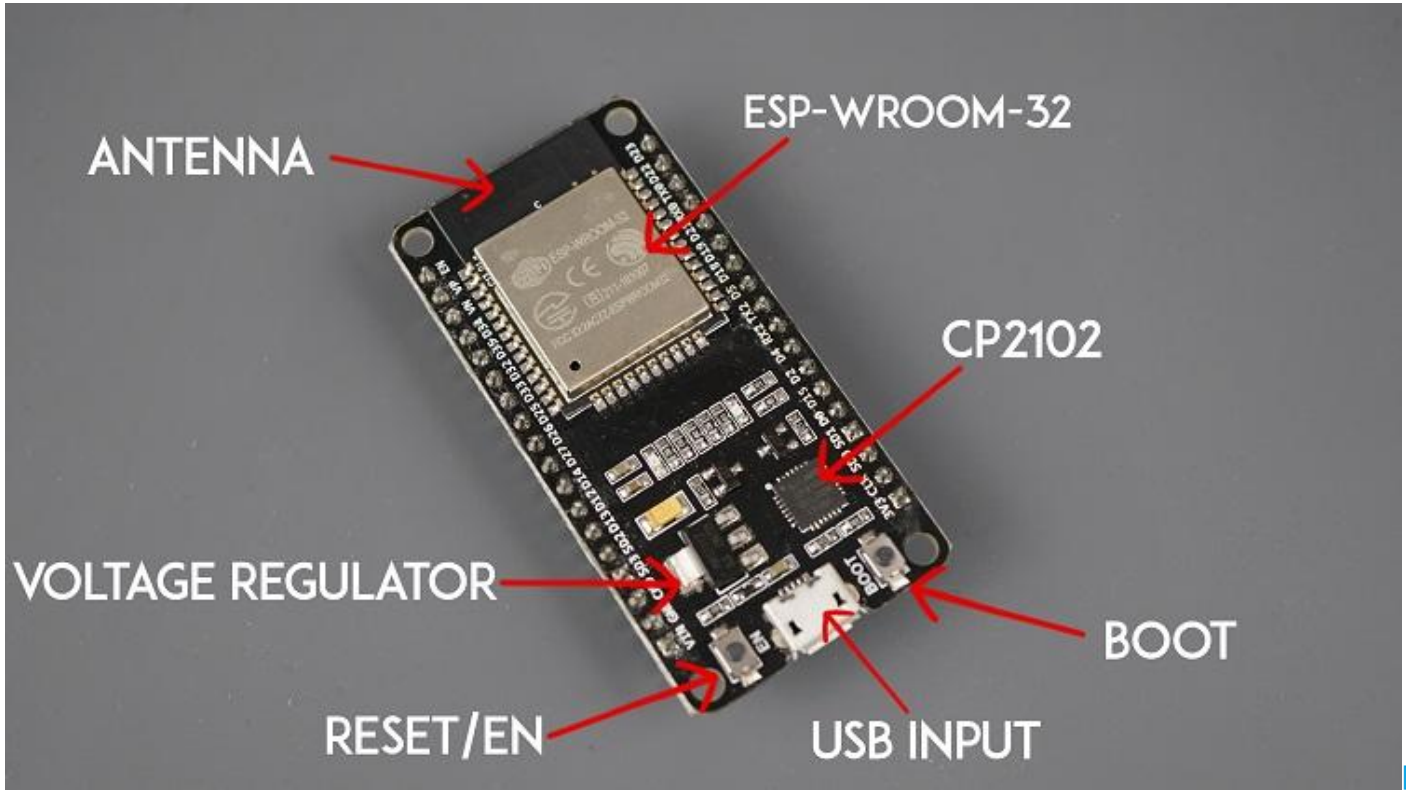
- História
  - ESP32-xx (2019-)
    - Lançamento de várias versões aprimoradas do ESP32 (datas de anúncio):
      - S2 (2019) – Single core, Xtensa LX7, Sem bluetooth, apenas WifFi (mais barato).
      - C3 (2020) – Single core, RISC-V, eficiente e barato.
      - S3 (2020) – Dual core, Xtensa LX7, suporte para AI vector instructions, com Wifi e Bluetooth.
      - C6 (2021) – Single core, RISC-V, suporte ao WiFi 6, Thread e Matter e Zigbee (802.15.4).
      - H2 (2022) – Single core, RISC-V, suporte a Zigbee (802.15.4), Thread e Matter (mas sem wifi).
      - P4 (2023) – Dual core, RISC-V, sem rádio, suporte a interface gráfica avançada, 32 MB de RAM e IA.



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Plataforma ESP32

- Características:



[20]

Plataforma ESP32

- Características:

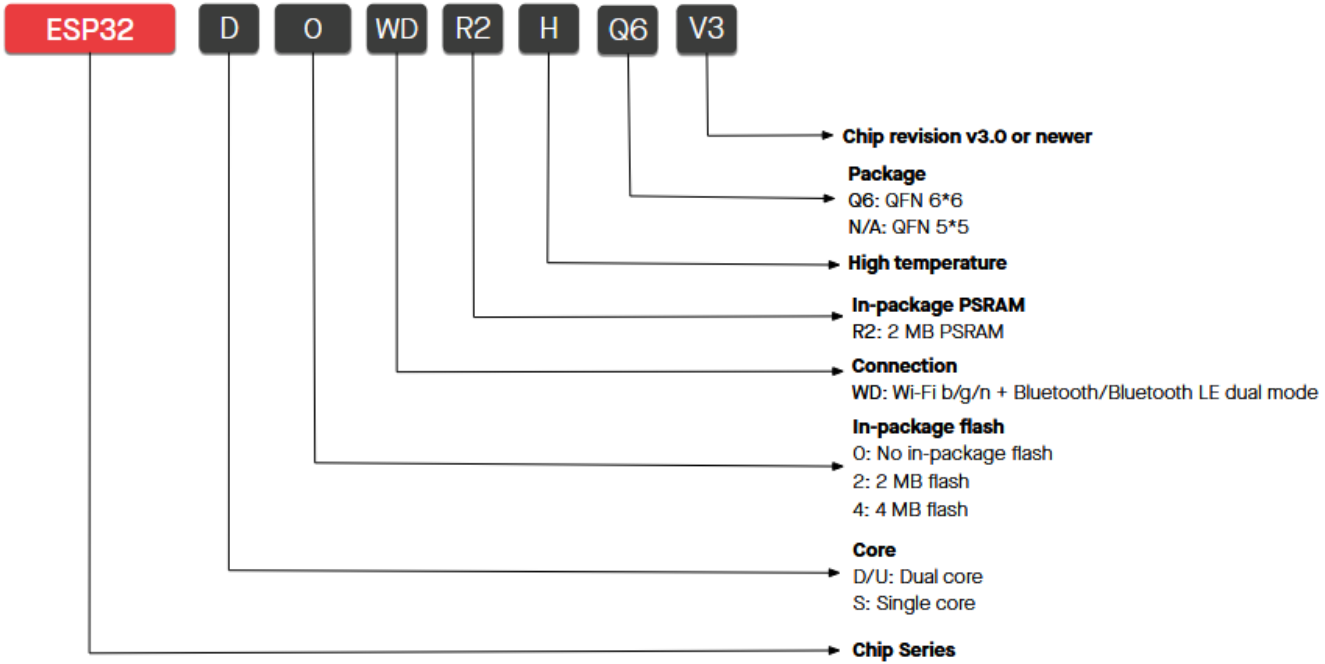
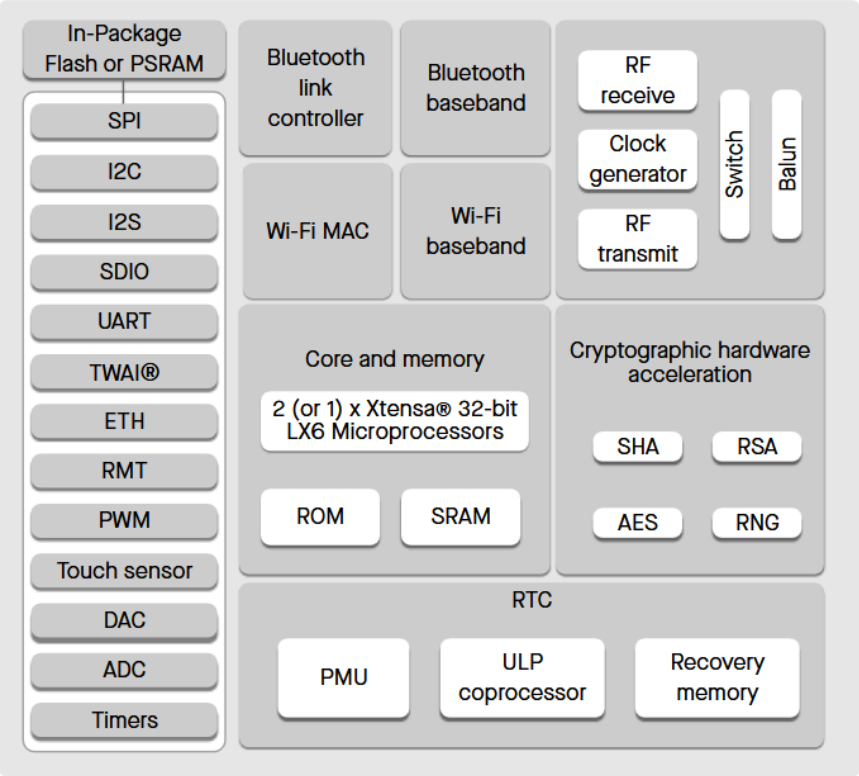


Figure 1-1. ESP32 Series Nomenclature

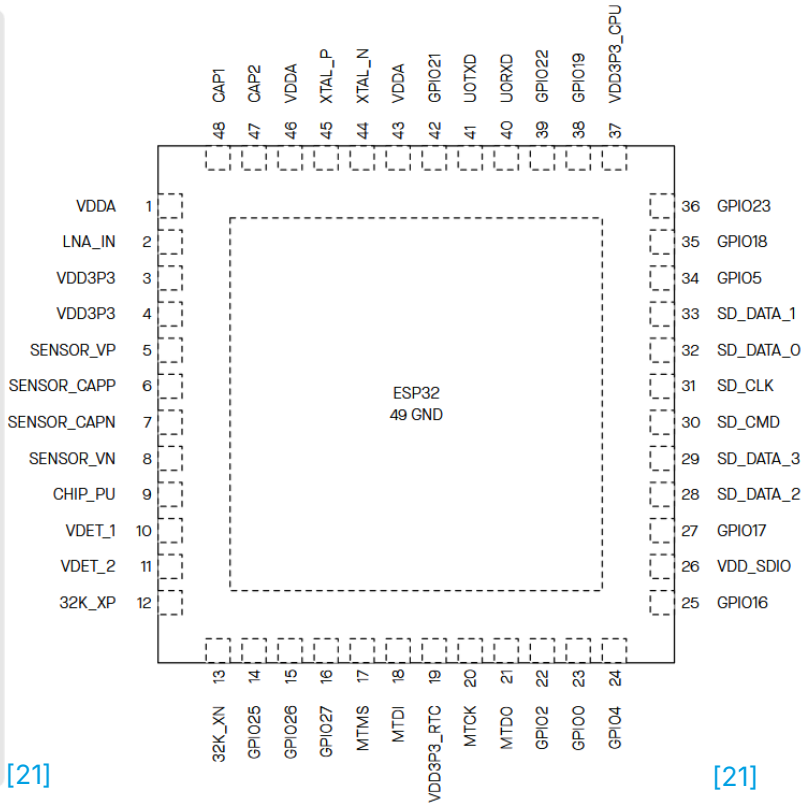
[21]

Plataforma ESP32

- Características:



[21]



[21]

Plataforma ESP32

- Características:
  - Tipo de arquitetura: 2 x Xtensa 32 bits RISC (depende do modelo).
  - Clock: 240 MHz (80 - 160 MHz, modo econômico).
  - Memória: 448 KB ROM, 520 KB SRAM, até 16 MB flash.
  - IO: 34 digitais (16 saídas PWM), 18 ADC (12 bits), 2 DAC (8 bits), 3 UART, 2 I^2C, 1 SPI.
  - Timers: 8 (64 bits, 4 por núcleo).
  - Criptografia por hardware (suporte a AES, SHA e RSA).
  - FreeRTOS (Sistema operacional em tempo real) integrado ao ESP-IDF.
  - Suporte a OTA (Over-The-Air Updates).
  - Protocolo próprio de comunicação (ESP-NOW).

Plataforma ESP32

- Características:
  - Mapeamento da memória.
  - Exemplo:
    - Endereço base GPIO = 0x3FF4_4000
    - Endereço base + 20 = porta como saída
    - Endereço base + 8 = “Setar” pino
    - Endereço base + C = “Resetar” pino.

Category	Target	Start Address	End Address	Size
Embedded Memory	Internal ROM 0	0x4000_0000	0x4005_FFFF	384 KB
	Internal ROM 1	0x3FF9_0000	0x3FF9_FFFF	64 KB
	Internal SRAM 0	0x4007_0000	0x4009_FFFF	192 KB
	Internal SRAM 1	0x3FFE_0000	0x3FFF_FFFF	128 KB
		0x400A_0000	0x400B_FFFF	
	Internal SRAM 2	0x3FFA_E000	0x3FFD_FFFF	200 KB
	RTC FAST Memory	0x3FF8_0000	0x3FF8_1FFF	8 KB
		0x400C_0000	0x400C_1FFF	
	RTC SLOW Memory	0x5000_0000	0x5000_1FFF	8 KB
External Memory	External Flash	0x3F40_0000	0x3F7F_FFFF	4 MB
		0x400C_2000	0x40BF_FFFF	11 MB+248 KB
	External RAM	0x3F80_0000	0x3FBF_FFFF	4 MB
Peripheral	DPort Register	0x3FF0_0000	0x3FF0_0FFF	4 KB
	AES Accelerator	0x3FF0_1000	0x3FF0_1FFF	4 KB
	RSA Accelerator	0x3FF0_2000	0x3FF0_2FFF	4 KB
	SHA Accelerator	0x3FF0_3000	0x3FF0_3FFF	4 KB
	Secure Boot	0x3FF0_4000	0x3FF0_4FFF	4 KB
	Cache MMU Table	0x3FF1_0000	0x3FF1_3FFF	16 KB
	PID Controller	0x3FF1_F000	0x3FF1_FFFF	4 KB
	UART0	0x3FF4_0000	0x3FF4_0FFF	4 KB
	SPI1	0x3FF4_2000	0x3FF4_2FFF	4 KB
	SPI0	0x3FF4_3000	0x3FF4_3FFF	4 KB
	GPIO	0x3FF4_4000	0x3FF4_4FFF	4 KB
	RTC	0x3FF4_8000	0x3FF4_8FFF	4 KB
	IO MUX	0x3FF4_9000	0x3FF4_9FFF	4 KB
	SDIO Slave	0x3FF4_B000	0x3FF4_BFFF	4 KB
	UDMA1	0x3FF4_C000	0x3FF4_CFFF	4 KB
	I2S0	0x3FF4_F000	0x3FF4_FFFF	4 KB
	UART1	0x3FF5_0000	0x3FF5_0FFF	4 KB
	I2C0	0x3FF5_3000	0x3FF5_3FFF	4 KB
	UDMA0	0x3FF5_4000	0x3FF5_4FFF	4 KB
	SDIO Slave	0x3FF5_5000	0x3FF5_5FFF	4 KB
	RMT	0x3FF5_6000	0x3FF5_6FFF	4 KB
	PCNT	0x3FF5_7000	0x3FF5_7FFF	4 KB
	SDIO Slave	0x3FF5_8000	0x3FF5_8FFF	4 KB
	LED PWM	0x3FF5_9000	0x3FF5_9FFF	4 KB
	eFuse Controller	0x3FF5_A000	0x3FF5_AFFF	4 KB
	Flash Encryption	0x3FF5_B000	0x3FF5_BFFF	4 KB
	PWM0	0x3FF5_E000	0x3FF5_EFFF	4 KB
	TIMG0	0x3FF5_F000	0x3FF5_FFFF	4 KB
	TIMG1	0x3FF6_0000	0x3FF6_0FFF	4 KB
	SPI2	0x3FF6_4000	0x3FF6_4FFF	4 KB
	SPI3	0x3FF6_5000	0x3FF6_5FFF	4 KB

[21]

- Características:
  - Exemplo de instruções Assembly do ESP32 (Xtensa)

Table 9: Core Instruction Summary

Instruction Category	Instructions[1]	Reference
Load	L8UI, L16SI, L16UI, L32I, L32R	*Load Instructions* on page 56
Store	S8I, S16I, S32I	*Store Instructions* on page 58
Memory ordering	MEMW, EXTW	*Memory Access Ordering* on page 60
Jump, Call	CALL0, CALLX0, RET J, JX	*Jump and Call Instructions* on page 61
Conditional branch	BALL, BNALL, BANY, BNONE BBC, BBCI, BBS, BBSI BEQ, BEQI, BEQZ BNE, BNEI, BNEZ BGE, BGEI, BGEU, BGEUI, BGEZ BLT, BLTI, BLTU, BLTUI, BLTZ	*Conditional Branch Instructions* on page 62
Move	MOVI, MOVEQZ, MOVGEZ, MOVLTZ, MOVNEZ	*Move Instructions* on page 65
Arithmetic	ADDI, ADDMI, ADD, ADDX2, ADDX4, ADDX8, SUB, SUBX2, SUBX4, SUBX8, NEG, ABS, SALT, SALTU	*Arithmetic Instructions* on page 66
Bitwise logical	AND, OR, XOR	*Bitwise Logical Instructions* on page 68

[22]

Plataforma ESP32

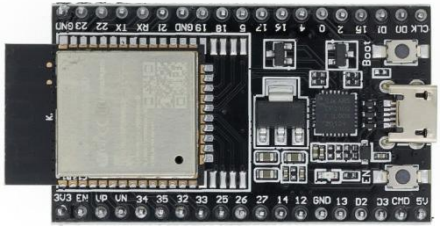
- Exemplo de código em assembly:

```
movi    a3, 4 /*Resultado para 1 << 2 Bitmask para GPIO2*/
l32r    a4, 0x3FF44020 /*Endereço do GPIO_ENABLE_W1TS_REG*/
s32i    a3, a4, 0      /*Escreve 1 no registrador para habilitar saída*/

l32r    a4, 0x3FF44008 /*Endereço do GPIO_OUT_W1TS_REG*/
s32i    a3, a4, 0          /*Escreve 1 no registrador para ligar o LED*/
```

Plataforma ESP32

- Modelos conhecidos:
  - Espressif: DevKitC V4, ESP32-PICO-Kit.
  - Ai-Thinker: NodeMCU-32S.
  - Wemos: LOLIN32.



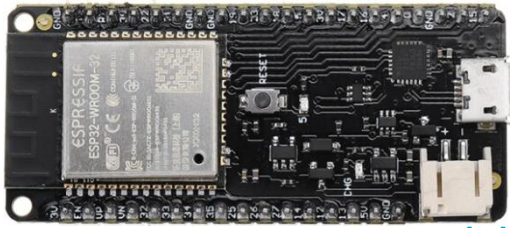
[23]



[24]



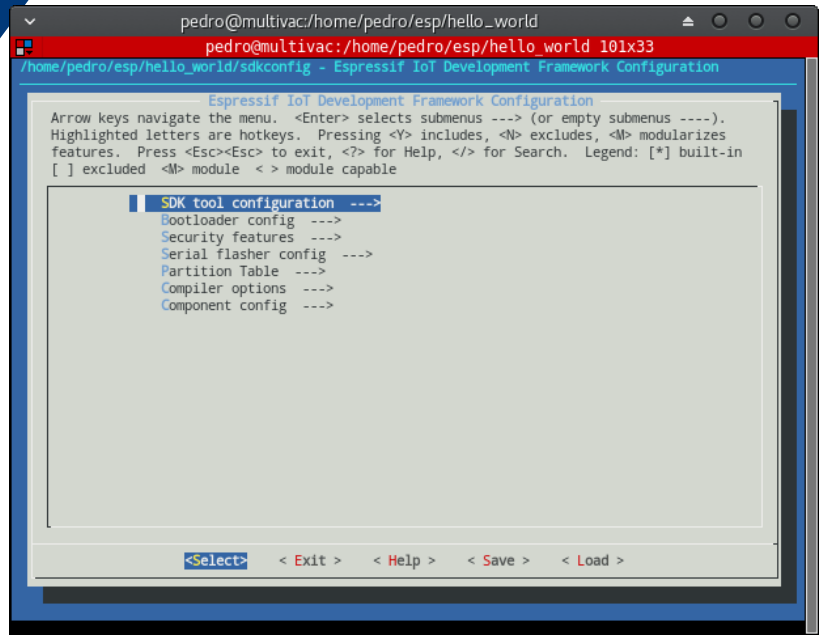
[25]



[26]

Plataforma ESP32

- Programação
  - IDF (Espressif):
    - SDK oficial da espressif
    - Conjunto completo de ferramentas, bibliotecas, e APIs para programar.
  - Alternativas
    - PlatformIO (VSCode) e Arduino IDE.



[27]

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IMD0904 –Plataforma de Hardware para Internet das Coisas

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